

Interpretable machine learning approach for exploring process-structure-property relationships in metal additive manufacturing

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ABSTRACT

Process-structure-property (PSP) relationships are critical to the optimization of manufacturing processes, but establishing these relationships typically involves time- and cost- consuming experiments, especially for additive manufacturing (AM) due to the large number of process parameters involved. In this study, we develop a novel and interpretable machine learning approach for predicting, optimizing, and expanding the process window of laser powder bed fusion (LPBF) while simultaneously establishing PSP relationships, using AlSi10Mg as an example. Our iterative, error-targeted method substantially decreases the amount of experimentation required. Gaussian process regression (GPR) was employed as the predictive model, incorporating multiple input variables (e.g., process parameters, relative density, melt pool morphology, cellular structure, and grain structure), for predicting three mechanical properties (i.e., yield strength, ultimate tensile strength, and % elongation). A comparison of model predictions and experimental data outside the training scope reveals that the prediction accuracy can be improved with higher dimensional inputs and further enhanced by a multi-output model that accounts for correlations between the different mechanical properties. Additionally, the GPR kernel's hyper-parameter for each input enables feature selection and model interpretability. The proposed approach can assist with finding the most critical variables affecting mechanical performance, establishing the PSP relationships of AM fabricated alloys, and providing guidance for tailoring the final properties. The methodology presented in this study can be applied to various AM techniques and materials to broaden the process window to achieve previously unattainable mechanical properties, as well as to gain a deeper understanding of the PSP relationships.

1. Introduction

Process-structure-property (PSP) relationships for metallic materials are important for informing material development and optimizing material manufacturing processes (forging, casting, etc.) to achieve desired materials properties. Establishing PSP relationships generally involves extensive, time-consuming experimental testing. Compared to conventional fabrication methods, additive manufacturing (AM) techniques pose a significantly greater challenge [1] due to their highly-dynamic processes (e.g., non-equilibrium solidification [2], cyclic thermal profiles [3], complex melt pool behaviors [4,5], etc.) and numerous process variables stemming from the layer-wise manufacturing processes, geometric design flexibility [6,7], material design flexibility [8,9] and highly tunable mechanical properties [10,11].

As an example, Fig. 1a shows the Ashby diagram of ultimate tensile strength (UTS) versus ductility for several common AM alloys including

Al-based alloys, Fe-based alloys, Ni-based alloys, Ti-based alloys, and high entropy alloys (HEA) processed using three popular metal AM techniques: laser powder bed fusion (LPBF), electron beam melting (EBM), and directed-energy deposition (DED). Notably, different process parameters (e.g., laser power and scanning speed as shown in Fig. 1b) can lead to significant variations in mechanical properties (e.g., about 200 MPa difference of UTS for LPBF AlSi10Mg in Fig. 1c) without noticeable processing defects (e.g., lack-of-fusion pores).

To define the process window and establish the PSP for metal AM, numerous methodologies have been developed either based on trial and error, physics-driven approaches, data-driven approaches, or combined approaches [12–15]. In the early stages of AM development, trial-and-error methods were commonly used because they can help to find suitable process parameters for fabricating specific materials with less time and effort compared to high-fidelity computational modeling, with the drawback being that the PSP exploration is constrained by the

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time-intensive and costly experimentation and that the determined process parameters may be sub-optimal. Physics-driven approaches generally rely on detailed modeling of the AM process at different time or length scales, including microscale [16], mesoscale [4] and macro-scale [17]. They require theory of the fast solidification process, such as melt pool dynamics, and also temperature-dependent properties of materials, such as thermal conductivity and reflectivity of the powder at high temperature, which are difficult to measure accurately. So, various simplifications are commonly introduced in physics-based models, which is regarded as one of the limitations compared to other approaches. And while the results from physics-driven approaches are generally interpretable in terms of how the various process parameters control the structure and properties, model simplifications can also possibly hinder the quantitative establishment of PSP relationships for metal AM. On the other hand, data-driven approaches are attractive for AM process development because they are inherently data-intensive manufacturing processes with numerous process parameters.

Over the past decade, the artificial intelligence, especially machine learning (ML), has become popular in the material science field to analyze data to solve various problems such as high-entropy alloy design [18], crystal structure determination [19], material reaction discovery [20], and so on. Furthermore, extensive effort has been made to employ machine learning algorithms in AM [21,22], including keyhole defect prediction [23], porosity prediction [24], microstructure prediction [25] and mechanical performance prediction [26]. However, even though the accuracy of ML predictions has been improved, the lack of interpretability of most “black box” ML algorithms limits their application on PSP exploration in AM since the details of how each process parameters affect the structure and properties is not readily accessible to the modeler [12]. Thus, there is a demand for interpretable ML models where the human users can understand the mechanisms behind the prediction model. An example of an interpretable ML model is where feature importance can be extracted and used to represent the influence of different variables on a prediction [27]. Additionally, the rapid development of AM printers with enhanced performance ranges (e.g., increasing laser power in LPBF from approximately one hundred watts a decade ago to over a kilowatt today) leads to broader possible process

windows within which different microstructures, crystallographic textures, and mechanical properties can be achieved. These recent developments have highlighted an urgent need for guidance on process parameter selection within an expanding process window by developing PSP relationships with more “process” possibilities using minimal experimentation, rather than conducting another exhaustive series of experiments within each new process window scope [28,29]. Therefore, to overcome these obstacles, an effective and physically interpretable ML approach designed for AM metallic material process window exploration and PSP relationship establishment would be attractive.

Among various machine learning and statistical analysis approaches, Gaussian process regression (GPR) has emerged as a valuable tool for exploration and exploitation scenarios in various domains, including battery [30], hydrology [31], wind energy [32], and other industrial fields [33]. GPR is popular because it has been demonstrated to exhibit a relatively high prediction accuracy and stability for small-dataset cases (also see Fig. S1) compared to other techniques including artificial neural network (ANN) and support vector machine (SVM) [34,35]. Even though there have been a few studies on the applications of ANN in AM field and achieved desirable prediction performance [36], the model performance can be very sensitive to the selection of hidden layer nodes while the GPR exhibit a more stable prediction accuracy in small-dataset scenarios [35]. Thus, this advantage of GPR is ideal for determining AM process windows while minimizing the cost of experiments. Furthermore, as a kernel-based surrogate model, GPR can employ different kernels for different types of datasets to capture the relationship between training data points and make predictions for unknown points. The ARD kernel, as mentioned in Section 2.4, provides the GPR model with the capability of feature/input selection, facilitating the determination of the relevance/importance of each input to output, which can be utilized to interpret the PSP relationships in AM processes. While there are some on-going developments and studies on how to achieve interpretability of traditionally non-interpretable neural network models by implementing several advanced techniques designed to extract, analyze, and visualize the inner workings and decision-making processes, these methods can significantly increase the computational burden. Thus, compared to other machine learning algorithms such as

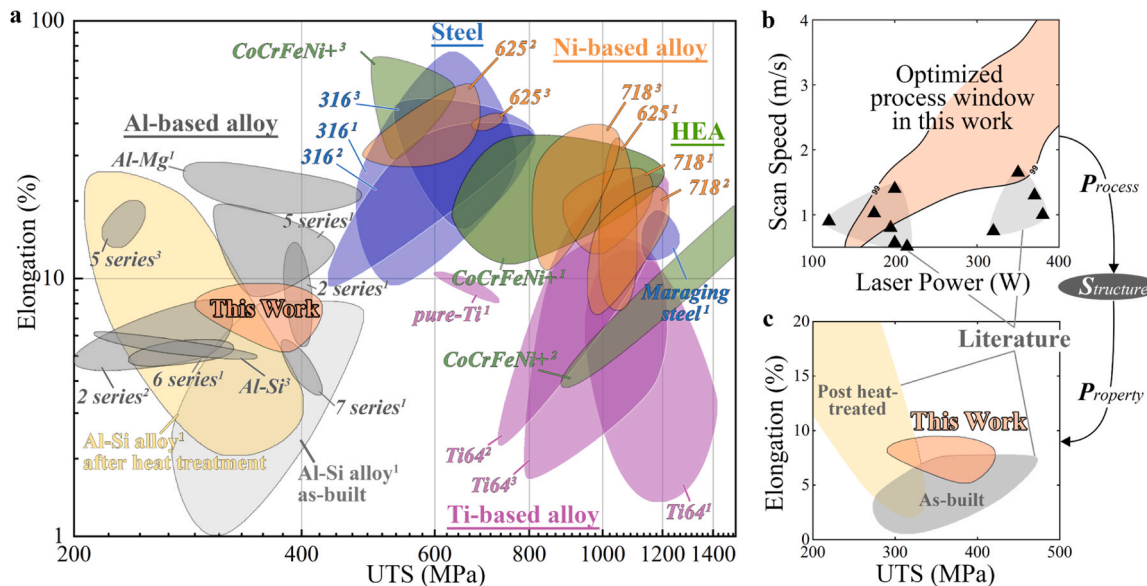


Fig. 1. Expanded process windows yield broader property ranges. a, Ashby plot showing the ultimate tensile strength (UTS) as a function of % elongation for commonly used metallic materials (as-built condition unless specified) fabricated via different additive manufacturing techniques. Superscript numbers correspond to: 1 = laser powder bed fusion; 2 = electron beam melting; 3 = directed-energy deposition. b, c, Optimized process window defined in this work (targeting on relative density > 99 %) for LPBF AlSi10Mg is marked as pink area. The reported process parameters for LPBF Al-Si alloys from literature is marked separately as grey color, demonstrating the larger process window enables the as-built material to achieve previously attainable mechanical properties, even with post-heat treatment (marked as yellow area).

ANNs, the GPR model stands out among numerous machine learning algorithms due to its inherent interpretability, which does not require such extensive computational resources and extra complex model structure. Moreover, the unique feature of error prediction of output results makes GPR suitable for creating iterative prediction-experiment loops to explore and expand process windows for AM metallic materials with minimal experimentation.

In this work, we aimed to develop a novel GPR model to define a high-density optimized process window for LPBF AlSi10Mg (one of the most widely used alloys for AM and can act as a simple model material for our investigation in this work) and then introduce a novel approach to efficiently explore an unknown process window area with higher laser power capacity through an iterative prediction-experiment loop while establishing the PSP relationship simultaneously, as shown in Fig. 2. To achieve this, we employed a GPR model with high-dimensional inputs by involving both process information as well as structural information and targeting mechanical performance as our predictive output. Our robust GPR prediction model is shown to be suitable for out-of-training-scope prediction, adapting to process window exploration in an upgraded power capacity scenario in LPBF. In the meantime, the PSP relationship of LPBF AlSi10Mg can be readily understood and established via an interpretable GPR model during prediction, utilizing the hyper-parameters in the ARD kernel. This new process window exploration and PSP relationship development approach can be used for other metals and alloys in the AM field to fully discover the potential of various manufacturing process parameters for tailorable mechanical performance, microstructure design, process certification, product qualification, etc.

2. Method

2.1. Material and LPBF process

Prealloyed AlSi10Mg powder (chemical composition in Table S1) with a diameter between 20 and 63 μm was acquired from LPW Technology Ltd (United Kingdom) for this study. The fabrication of all specimens in this work was conducted on an SLM Solutions 125^{HL} printer (Germany) equipped with a fiber laser with a maximum power of 400 W, under an atmosphere filled by high purity argon gas with

controlled oxygen content below 0.02 % to minimize oxidation. The build plate was preheated to 200°C to relieve residual stress in the as-built parts [37] and the bidirectional laser scanning strategy was used a 90-degree angle rotation increment between layers. Two major process parameters (i.e., laser powder and scan speed) were changed as variables to define the investigated process window in this study, as listed in Table S3, while layer thickness and hatching space were kept constant at 30 and 100 μm , respectively.

2.2. Mechanical property measurement

Different types of specimens were fabricated with a range of parameters for obtaining various properties of LPBF AlSi10Mg. Firstly, parameter combinations shown in Fig. S1a were used to fabricate 5×5×5 mm³ cubes for relative density measurement. Density measurement were conducted using an A&D Weighing HR-250A balance based on Archimedes method and 2.67 g/cm³ was used as the theoretical density of AlSi10Mg. Secondly, parameters shown in Table S3 were used to fabricate dog-bone tensile specimens with a cross-section of 2×4 mm and a gauge length of 18 mm. An Instron 3369 machine (United Kingdom) was employed for tensile testing in general accordance with the ASTM E8 standard [38] at a displacement rate of 1 mm/s at room temperature. The obtained stress-strain curves for the different samples were used for acquiring the corresponding mechanical properties including yield strength (0.2 % offset), ultimate tensile strength and % elongation (Table S3). A 12.5 mm gauge length extensometer was used for tensile strain measurement. All measurements (i.e., density, tensile properties) were averaged from at least three specimens for each combination of process parameters.

2.3. Mesostructure and microstructure characterization

The melt pool cross-section morphology was simulated by a thermal profile of AlSi10Mg during LPBF, which is determined using a simplified finite element analysis (FEA) thermal model developed in COMSOLTM Multiphysics software suite. Heat input from laser irradiation is expressed as [39,40]

$$Q(x, y, z) = \frac{2AP}{\pi\delta R^2} e^{-\frac{2(x^2+y^2)}{R^2}} e^{-\frac{|z|}{\delta}} \quad (1)$$

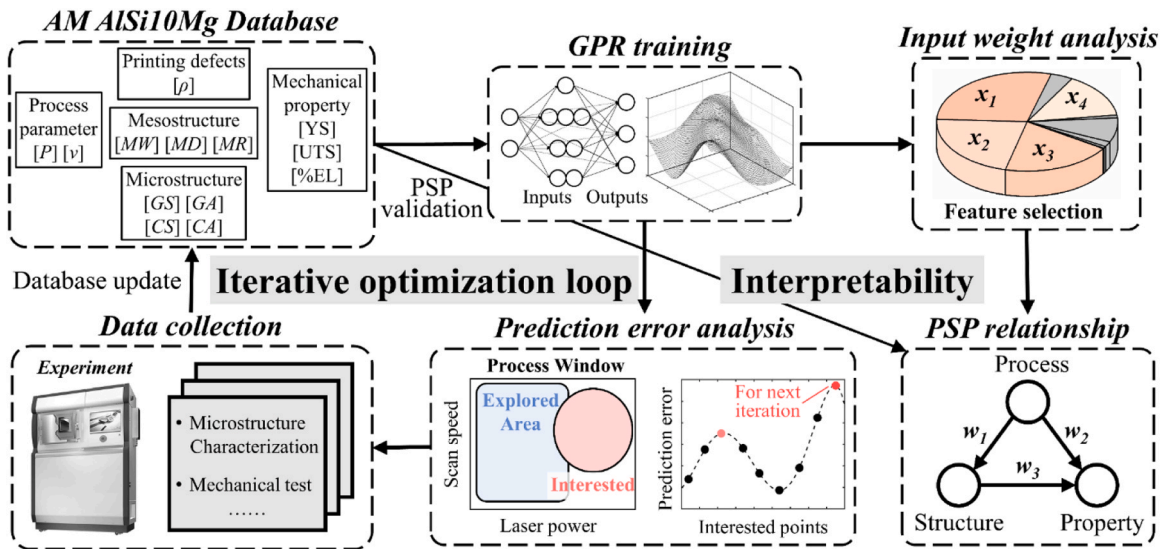


Fig. 2. Overview of model framework introduced in this work. We developed a GPR model to analyze obtained data from LPBF fabricated AlSi10Mg and make predictions of mechanical properties with unexplored process parameters along with the prediction error, which is an intrinsic advantage of the GPR approach. Based on the prediction error, the best candidates for the next step of experimentation can be identified and fabricated to provide more data feedback to the database, further improve the prediction accuracy, and explore unknown process window areas. Additionally, the GPR model will generate corresponding weights for each input variable to reflect their importance in the prediction model, which help to develop the PSP relationships in metal AM processes.

where Q represents the volumetric heat flux (W/m^2), A the laser absorption coefficient, P the laser power, R the laser beam radius, δ the laser penetration depth, $|z|$ the z -coordinate absolute value, and x and y the laser spot x - and y -coordinates, respectively. For AlSi10Mg, the density is $2.67 \text{ g}/\text{cm}^3$ while specific heat capacity and thermal conductivity are $800 \text{ J}/(\text{KgK})$ and $160 \text{ W}/(\text{mK})$, respectively [41]. Prior to simulation, single-scan tracks were printed on an AlSi10Mg sample using the SLM 125^{HL} machine. Parameters for P , R , and the x - and y -coordinates (laser scan speed) used in simulation were same as applied for single-scan track printing. Meanwhile, to determine the penetration depth δ and absorption coefficient A , an iterative approach was used. In the simulation, δ and A were manipulated until the simulated melt pool aligned with the observed melt pool. This iterative process was only conducted for $P = 200 \text{ W}$ and $v = 1400 \text{ mm}/\text{s}$, while three other conditions of process parameters were used to validate the obtained δ and A values. The values of δ and A were optimized to $6 \mu\text{m}$ and 0.12 , respectively, and all other parameters except for δ and A were kept constant. Once δ and A were determined, the impact of varying laser power and scan speed on melt pool geometry was investigated. A build platform preheat temperature of 200°C was used as the starting condition with a convective heat flux heat transfer coefficient of $10 \text{ W}/(\text{m}^2\text{K})$ and a surrounding temperature of 20°C . Surface to ambient radiation was also considered with surface emissivity of 0.3 . The profile shown in Fig. S2 represents the simulated single scanning track using $P = 200 \text{ W}$ and $1400 \text{ mm}/\text{s}$ as the scanning speed. The melting temperature of AlSi10Mg (570°C) was marked out and determined as the edge of the melt pool, and the overlap of simulated melt pool and observed melt pool from single scan track validates the model and iterative process discussed in the paragraph above. Even though the simplified model cannot achieve a high accuracy without considering the powder bed, it can reflect a rational trend of how the scan speed and laser power will affect the melt pool dimensions. The simulated melt pool width (MW), depth (MD) and its ratio (MR) are listed in Appendix B, and used as input variables in the training model.

An AURIGA CrossBeam Workstation (Carl Zeiss, Germany) and a Nova NanoSEM 450 filed-emission scanning electron microscope (SEM, FEI, USA) were used for characterizations of grain microstructure and cellular microstructure of LPBF AlSi10Mg, respectively. The grain microstructure was characterized by electron back-scattered diffraction (EBSD) with a fixed step size of $0.5 \mu\text{m}$, $12,000\times$ magnification, a working distance of 5.6 mm , an acceleration voltage of 5 kV , and a spot size of 3.0 . All specimens were progressively ground to a P4000 SiC paper finish followed by progressive polishing down to a $1 \mu\text{m}$ diamond solution. Then, specimens for EBSD grain structure characterization were final polished using a $0.04 \mu\text{m}$ OP-U non-drying colloidal silica suspension while specimens for SEM imaging were etched by Weck's reagent (4 g KMnO_4 , 1 g NaOH and 100 ML distilled water) to reveal the cellular microstructure. The size and aspect ratio of grain microstructure was determined by grain area and equivalent grain diameter method respectively via Oxford Instruments HKL Channel 5 software (United Kingdom), and the size and aspect ratio of LPBF AlSi10Mg cellular microstructure was analyzed and determined by the image processing technique developed in our previous work [42]. The averaged data of grain size (GS), grain aspect ratio (GA), cellular size (CS), cellular aspect ratio (CA), as inputs of the model in this work, for all samples fabricated with different process parameters can be found from Appendix B.

2.4. Gaussian process regression model

Gaussian process regression (also known as Kriging in statistics) is an interpolation method based on a Gaussian process where a collection of random variables has a joint Gaussian distribution within a continuous domain [43]. In this work, we employed both a classic single-output GPR model [43] and a multi-output GPR model developed by Chen et al. [44,45], the latter of which enables us to make the prediction on

different mechanical properties as multi-outputs simultaneously and which utilizes the relevance of all outputted mechanical properties to further increase the prediction accuracy. The classic single-output GPR prediction is jointly determined by its mean function and covariance function (also called "kernel"), the latter of which captures the relationship between different input points and can be optimized to make prediction for unknown points from the input training data. On the other hand, the multi-output GPR model utilizes a cross-covariance function, which describes how outputs are related to each other in addition to how they relate to the inputs. This model will not only capture the underlying output influence on each other, but also reduce the computational cost and significantly improve the prediction efficiency compared to single-output models. In this work, we applied an automatic relevance determination (ARD) kernel for all models because this kernel has different length-scales $\ell_1, \dots, \ell_D$ (i.e., hyperparameters) in its covariance function for each dimension, which can be used to determine how relevant an input is: if the length-scale is larger, the kernel for prediction will become more independent of the corresponding input. Thus, the inverse of the length-scale can reflect how much each input can affect the output prediction. Then, the weight (w) of each input variable in different GPR models can be calculated:

$$w_i = 100\% \times \frac{1/\ell_i}{\sum_{j=1}^D 1/\ell_j} \quad (2)$$

Moreover, the GPR algorithm operates on the principle that observations can be modeled as samples from a continuous probability Gaussian distribution, allowing it to naturally incorporate uncertainty in its predictions, which is denoted by "predictive uncertainty" in this work. On the other hand, the "prediction error" of the GPR model was defined by averaging the deviation of the predicted value from the experimental one:

$$Error = 100\% \times \frac{1}{n} \times \sum_{i=1}^n \frac{|x_{p,i} - x_{ex,i}|}{x_{ex,i}} \quad (3)$$

where n is the number of validating points while $x_{p,i}$ and $x_{ex,i}$ are the predictive and experimental values of the property, respectively.

In this study, the training dataset comprises a total of 42 data sets. Initially, a subset of these data sets is chosen to serve as the training dataset while others serve as validation dataset (as illustrated in Fig. 3a), followed by re-training after transferring specific points from the validation dataset to the training dataset, for the purpose of illustration of our proposed iterative prediction-experiment process (demonstrated in Section 3.1). Subsequently, all 42 data sets are used as training dataset for generating input weights and facilitating the exploration of PSP relationship (demonstrated in Section 3.2).

3. Results and discussion

3.1. Mechanical property prediction of Gaussian process regression model

To ensure the LPBF fabrication quality, addressing processing defects (e.g., porosity) in the fabricated parts is a primary objective during process optimization. We initially trained a GPR model to predict relative density by varying two key process parameters of LPBF: laser power and scan speed. Fig. 3a shows an optimized density map for LPBF AlSi10Mg as a function of these two parameters ($>99\%$ relative density area marked with contours). Based on predicted density map, we divided our process window into two subareas, where the red area with higher laser power capacity denoting unexplored area while the blue circles represent the initial training data points. It should be mentioned that parameters near two corners in Fig. 3a were excluded due to either low density resulting from lack of fusion or printing failures triggered by over-melting and melt pool instability. Extrapolation error (shown in Fig. S1) is a common issue with GPR models that often makes them

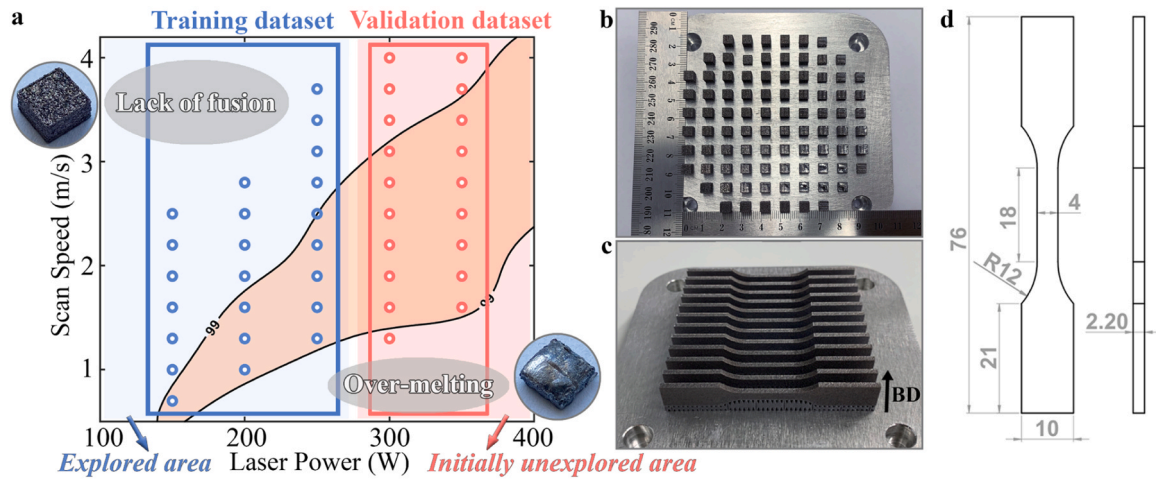


Fig. 3. Dataset selection and training scope definitions. a, A process window with a higher laser power capacity (≥ 300 W) is designated as an unexplored area and marked as red. Blue and red circles represent the training and validation dataset, respectively. Unsuitable process parameters failed to fabricate tensile test specimens due to either low fabrication quality (i.e. porosity) or printing failure issues such as key-hole due to over melting, as shown in the insets [46]. b,c, LPBF fabricated cubic samples and tensile test specimens for density measurement and tensile tests (build direction, i.e., BD, is marked in the figure), respectively. All combinations of process parameter were used to fabricate at least three samples/specimens. d, the schematic drawing of the tensile specimen used in this work.

unsuitable for out-of-training-scope prediction, such as in the context of an upgraded laser power capacity scenario in LPBF. To overcome this limitation and adapt the GPR model for continuous exploration of expanding process windows across all AM techniques, we employed multivariate GPR models to enhance mechanical property predictive performance by involving both process- and structure-related information. As schematically illustrated in Fig. 4, we considered different levels of structural characteristics of the fabricated material, including macrostructure (i.e., relative density, defects, etc.), mesostructure (i.e., melt pool structure), and microstructure (i.e., grain and cellular structure). Following this, a total of ten representative variables were selected in this work to characterize these structures as well as the LPBF process (i.e., laser power and scan speed) in the GPR model. They are cumulatively used as multivariate inputs of single-output GPR models to predict three mechanical properties separately, followed by a multi-output GPR model which can predict different mechanical properties simultaneously.

Yield strength, one of the most critical mechanical properties in

various engineering applications, was selected to demonstrate the outside-of-training-scope prediction performance of various GPR models for LPBF AlSi10Mg and how the performance could be enhanced. Firstly, the laser power (P) and scan speed (v) were used as two input variables for single-output GPR model training, which resulted in relatively large error bars for all validating points (Fig. 5a) due to the extrapolation error, as mentioned in Fig. S1. Including relative density (ρ) as the third input variable improved prediction performance (Fig. 5b), as physical density or printing defects can significantly impact the final mechanical performance of the fabricated parts, thus providing more robust structure-property relationship information for output prediction. Similarly, incorporating meso-structure information into the model, such as melt pool width (MW), melt pool depth (MD), melt pool depth/width ratio (MR), and micro-structure morphologies, such as grain size (GS), grain aspect ratio (GA), cellular size (CS), cellular aspect ratio (CA), further enhanced prediction performance outside training data scope (i.e., unexplored area in this case) with the prediction error decreasing from 12.4 % to 8.7 %, as shown in Fig. 5c. This is because the

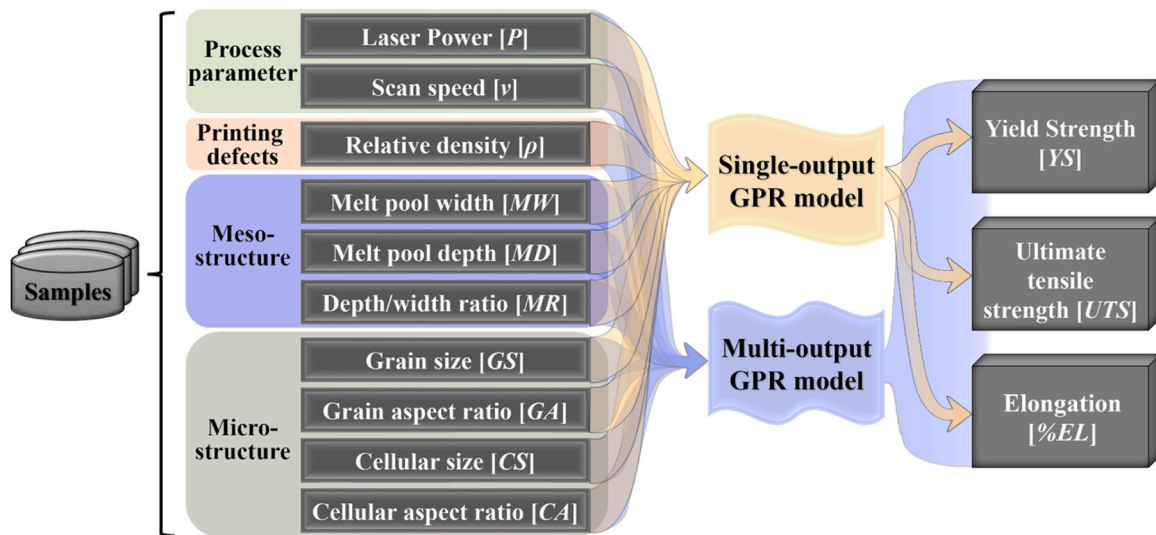


Fig. 4. Schematic of the GPR prediction models and variables. Different process parameters for sample fabrication and their corresponding physical & structural information are used as inputs, while the mechanical properties are used as outputs. Two different GPR models (single-output and multi-output) were employed in this study with the aim of comparing their predictive performance.

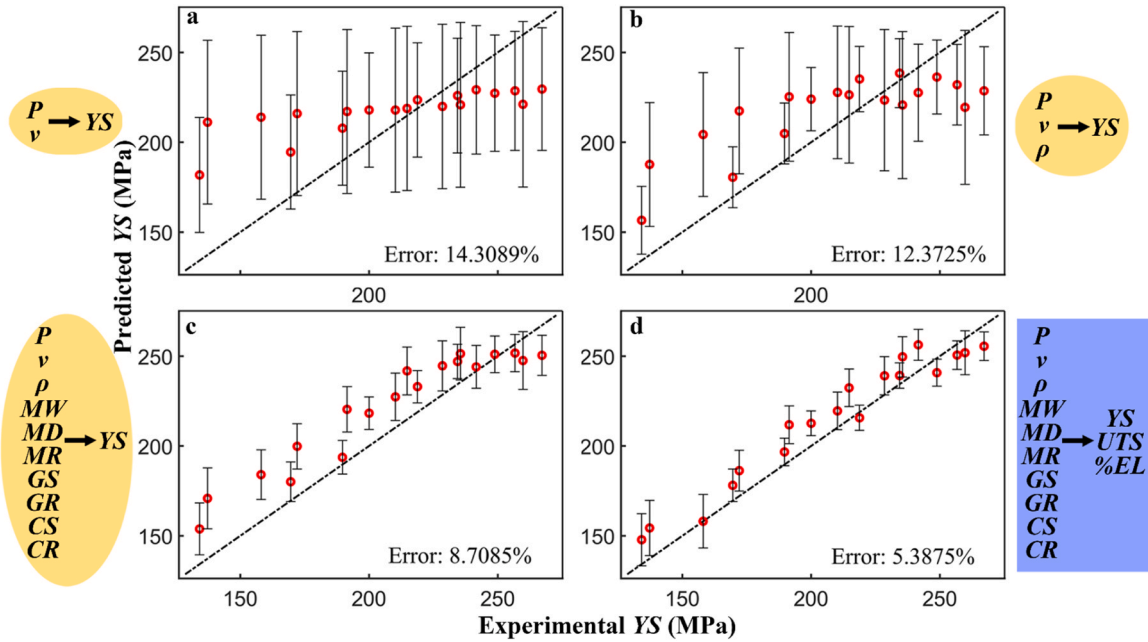


Fig. 5. Predictive performance of different dimensional GPR models. Red circle points in the validation dataset (Fig. 2a) show the predictive performance of our different GPR prediction models. a, b, and c are results from single-out model and d is the result from multi-output model. Error bars denote one standard deviation of the predicted value.

microstructure and mesostructure of AM metallic materials have been known as one of important factors affecting the mechanical properties [42,47,48]. By involving these structure variables into our model, they offer the potential to establish correlations of structure-property relationships and provide useful information for mechanical performance prediction. In contrast to the single-output models, multi-output GPR models using yield strength, ultimate tensile strength, and ductility at outputs exhibited substantially better prediction performance with reduced prediction error on yield strength (see Fig. 5d), which can be attributed to the involvement of the correlation relationships between the various mechanical properties. For instance, low yield to tensile ratio is related to large elongation for high-strength steel [49]. The multi-output GPR model can capture these latent correlations for specific material through training, thereby enhancing predictive accuracy. Another advantage of using a multi-output GPR model is the simultaneous prediction of three mechanical properties, yielding higher computational efficiency compared to single-output prediction model.

3.2. Enhanced prediction model accuracy via an iterative prediction-experiment process

To further improve the model prediction performance for the process optimization and prediction of an untrained area with higher laser power capacity (i.e., above 300 W as shown in Fig. 3a), we proposed the iterative process schematically described in Fig. 2. One of the outstanding features of GPR models is their ability to provide the deviation from predicted values (i.e., predictive uncertainty). For visualization and better illustration of how the large predictive uncertainty can be controlled, the single-output $P, v \rightarrow YS$ model was selected for illustration (Fig. 6a and b), where the two grey uncertainty surfaces sandwiching the predicted mean value surrogate surface represent one standard deviation. Given that process parameters directly impact solidification during LPBF and subsequently influence density, micro-/meso-structure and, consequently, the mechanical properties of the printed material, it is reasonable that the high-power area shows higher predictive uncertainty due to the lack of training points providing useful information to establish the correlation between process parameters and mechanical property. Then, the predictive uncertainty for all validation

data points were compared and the point with the highest predictive uncertainty (marked as a red pentagram in Fig. 6b) was selected and added to the training dataset for the next iteration, which markedly reduced the uncertainty near the new added point, thus proving the feasibility of the iterative process for efficient GPR model development. Although the prediction-experiment iteration was illustrated here for yield strength based on a ten-dimensional input model, it is also applicable for rapid process window exploitation targeting on density (i.e., $P, v \rightarrow \rho$ model), which is an efficient way to reduce the extrapolation error in the GPR prediction mentioned above. In addition, the iterative prediction process was also applied on our multi-output GPR model and the corresponding overall prediction errors on YS at each iteration is shown in Fig. 6c. In this way, the prediction model provides guidance to most efficiently explore the process window with the least possible experimental effort for any metal AM technique. In this study, a broader process window than those previously reported in the literature was quickly found, which enabled us to fabricate samples with previously unattainable mechanical performance, as demonstrated in Fig. 1b and c, which was not easily achievable by post-heat treatment. For instance, the desirable combination of high tensile strength (376 MPa) and large elongation (9.3 %) can be obtained by process parameter of 300 W laser power and 2200 mm/s scan speed in this work.

3.3. Significance of various model input variables in impacting the mechanical properties

As described in the Section 2.4, the length-scale parameters ℓ_j in the ARD kernel provides the GPR model with the capability of feature/input selection. We trained three GPR models to predict the mechanical properties of LPBF fabricated AlSi10Mg utilizing all 42 experimental data points as the training dataset (i.e., both red and blue circles in Fig. 3a). The corresponding single-output surrogate surfaces, with P, v plotted on the x, y coordinate axes, respectively, have been shown in Fig. 7. It is worth noting that the predicted mechanical property maps as shown in Fig. 7 were based on ten input variables while only two variables (i.e., process parameters) were used here merely for visualization purposes. Given the impracticality of simultaneously visualizing the impact of all input variables in a three-dimensional figure, we utilized

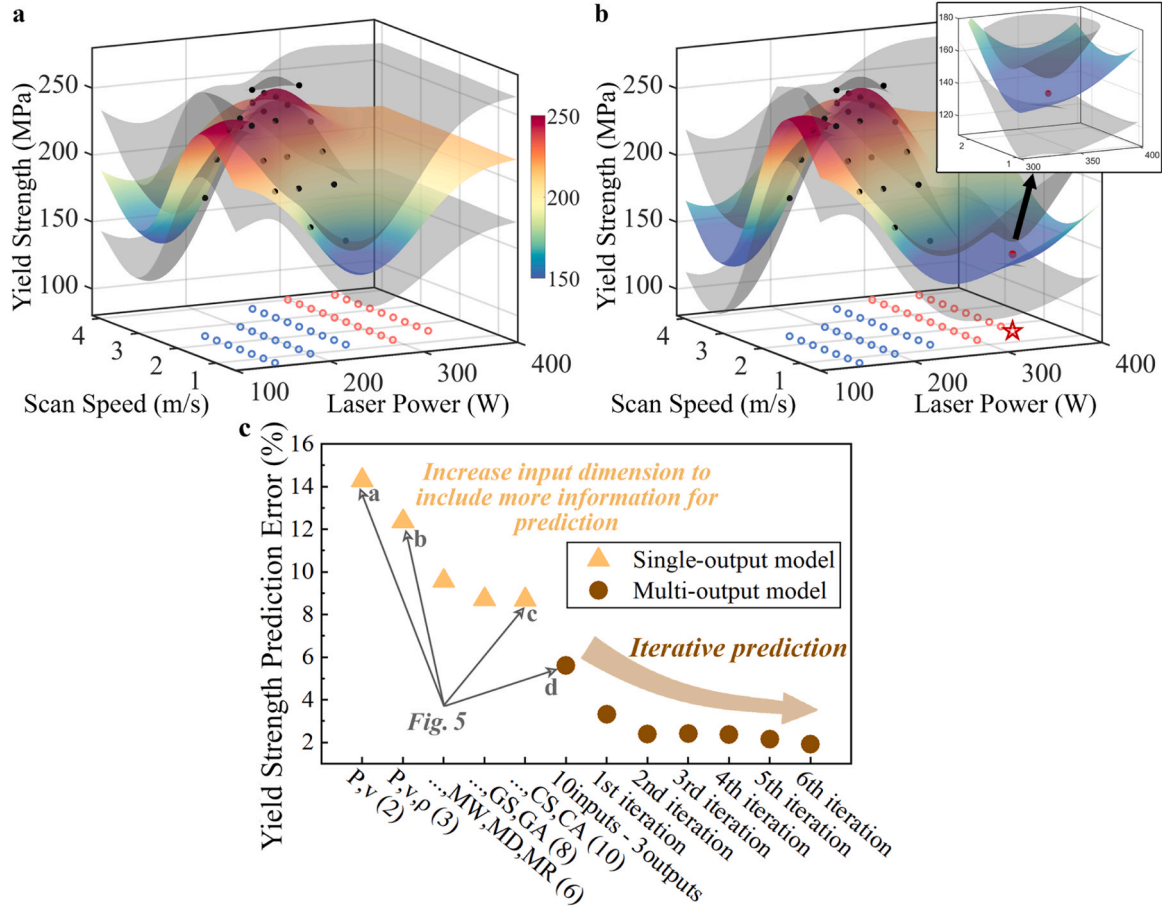


Fig. 6. Out-of-training-scope prediction and iterative experimental exploration loop targeting yield strength. **a**, Predicted yield strength (YS) surrogate surface via single-output GPR model with P, v as inputs. The top and bottom grey surfaces represent one standard deviation of the predicted value generated from GPR to denote the predictive uncertainty. **b**, New predicted model of YS after the 1st iteration, by adding one more data point (marked as red pentagram point) into training dataset. Inset in **b** shows the refined predictive uncertainty via iteration. **c**, yield strength (YS) prediction error for each GPR model: yellow triangle points represent the single-output model with different dimensional inputs; brown circle points represent the initial multi-output model as well as after up to six prediction-experiment iterations.

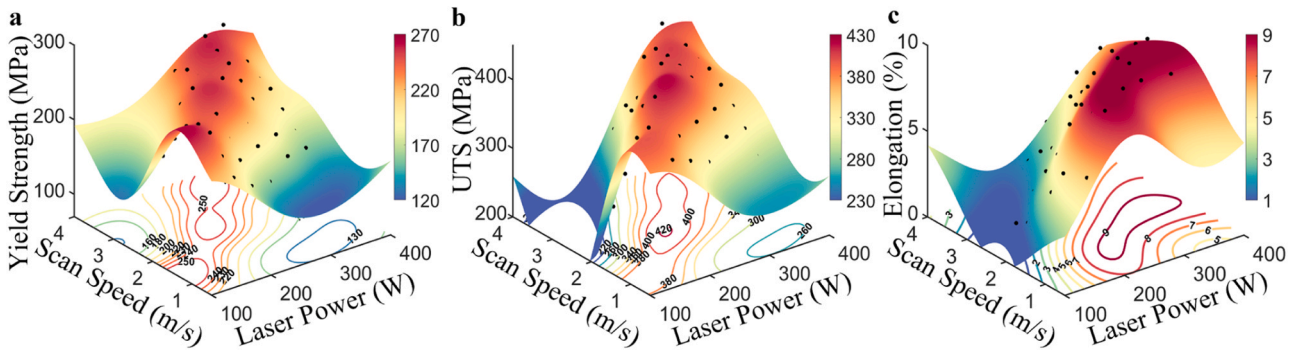


Fig. 7. Predictive surrogate surface generated from single-output GPR models for three different mechanical properties of LPBF AlSi10Mg: **a**, Yield strength; **b**, Ultimate tensile strength; **c**, % Elongation.

the inherent interpretability of GPR model to analyze and understand the influences of these variables on mechanical properties in this study.

For each GPR model, we calculated the percentage weight of each input based on its length-scale using Eq. (2) with the results shown in Fig. 8a. Fig. 8a thus illustrates the relative influence of each corresponding input variable on the predictive output, acting as an efficient factor for feature selection to create an accurate GPR model using fewer inputs. Intuitively, a low weight suggests that the changes in the corresponding input variables do not directly affect the output results, such

as the grain aspect ratio has no apparent relationship to the yield strength of the LPBF fabricated AlSi10Mg as shown in Fig. 8b. While the average aspect ratio derived from 2D EBSD image can indicate the grain orientation, which will impact the mechanical properties of crystalline materials [50,51], our observations indicate a minimal influence of GA in this context. This is mainly because the scan strategy (i.e., 90-degree angle layer increment rotation) applied in this study remain same for all fabricated samples as mentioned in Section 2.1, resulting in a similar grain orientation or texture across all fabricated material (EBSD images

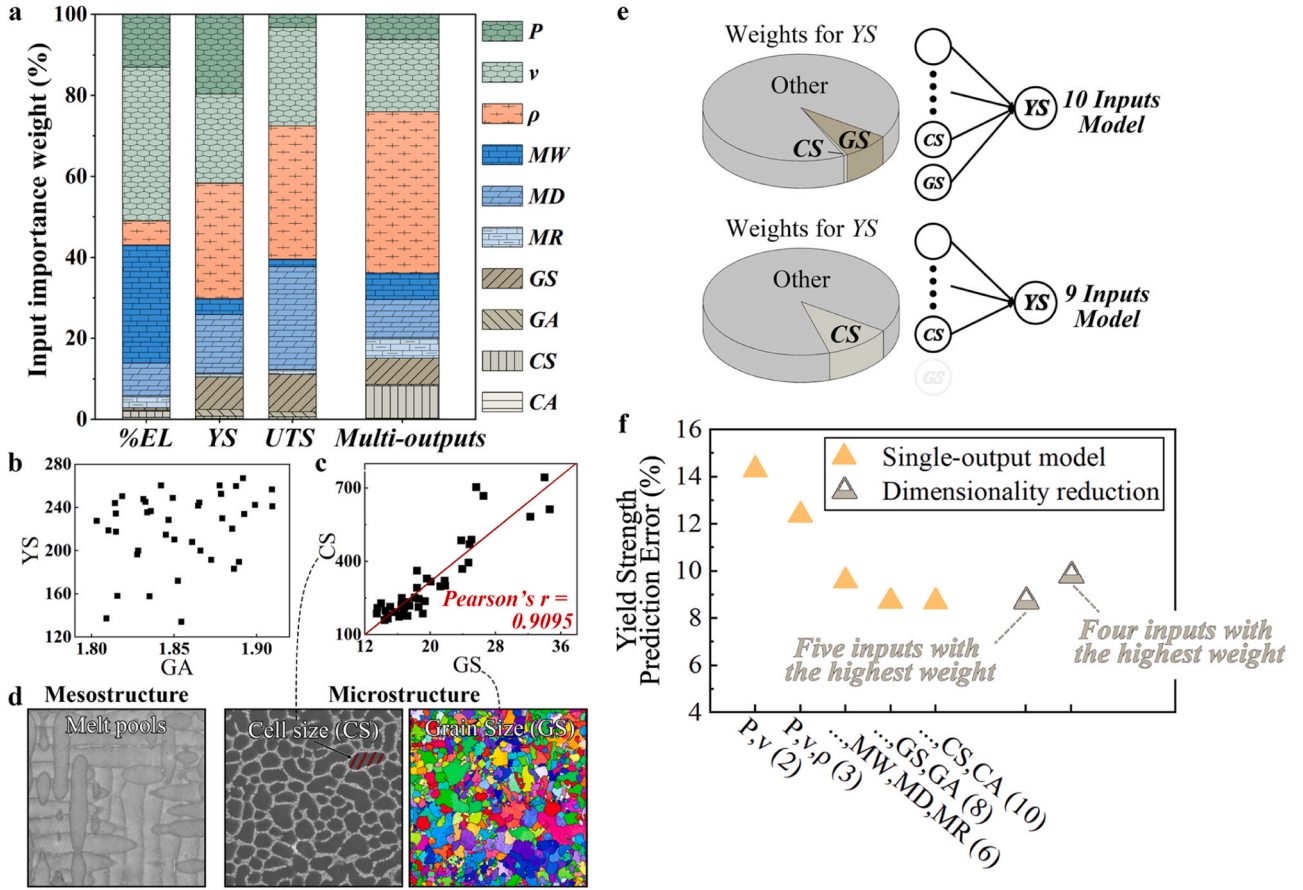


Fig. 8. Weight of inputs generated from GPR prediction model. **a**, Weights of each input for the mechanical property predictions obtained from different GPR models (see Table S2 for detailed weights). **b**, Grain aspect ratio has minor impact on the yield strength of LPBF AlSi10Mg, showing no correlations between input and output. **c**, Melt pool depth and melt pool width have a strong linear correlation. **d**, Optical microscope image shows a melt pool mesostructure characterized by a 90-degree rotation scanning strategy pattern in LPBF AlSi10Mg; SEM image shows a distinctive cellular structure, where boundaries are enriched with Si-rich phase and α -Al phase as matrix; EBSD image shows a fine, equiaxial grain structures typical of LPBF AlSi10Mg. **e**, Two training models with corresponding weights of CS. Comparison of training models, with and without GS as one of the inputs, reveals the CS has been dewighted due to the correlation to GS in the first model. **f**, Prediction error remains accurate after input dimensionality reduction based on the weight of each input variable.

available in Table S3). The critical role of scan strategy on texture development has also been reported in other LPBF fabricated metals and alloys [52]. Thus, identical scan strategy underscores the importance of the grain aspect ratio our scenario. On the other hand, it should not be automatically inferred that all variables with small weights will have no relationship to the corresponding mechanical performance. Because the small weight assigned to a certain input parameter may also be attributed to its linear correlation with other input parameters. One example is the correlation between cellular size and grain size as shown in Fig. 8c, because both variables will be affected similarly by the cooling rate during solidification. More specifically, during the initial stage of the solidification process, atoms in the liquid start to arrange themselves into a crystalline order, forming small clusters known as nuclei. When the liquid metallic material cools down rapidly, the atoms have less time to migrate and attach to the growing grain, resulting in larger number of nuclei forming and less time for each nucleus to grow, leading to finer grain structure. And our result is consistent with the model [53]: $\bar{d} = \bar{a} + D \cdot \Delta T_n / (\nu \cdot Q)$, where $\bar{d}$ is the average grain size, $\bar{a}$ is the spatial distance, D is the diffusion coefficient, ΔT_n is the cooling rate, ν is the growth velocity, and Q is the growth restriction factor. Similarly, the typical cellular structure of LPBF AlSi10Mg is formed by Si-rich phase precipitation [54]. The faster cooling rate will limit the diffusion of solute atoms (i.e., silicon atoms), preventing them from segregating into larger/coarser precipitates. Thus, both the cellular structure [42] and grain size [55] of LPBF AlSi10Mg tend to be finer with higher cooling

rate. From the perspective of the training model, when one variable becomes more advantageous in facilitating precise predictions, the weight of the other variable diminishes correspondingly. This can be attributed to their inherent correlation, which diminishes the benefit of additional information provided by the latter variable. Even though CS can be dewighted by the correlation with GS, the influence of CS can be revealed when removing GS from the training input variables, as shown in Fig. 8e. Melt pool depth and width also have such scenario since their tendencies to increase or decrease are similarly influenced by the changes of laser intensity. Therefore, the weights generated along with the GPR model can be used for feature selection and subsequently input dimensionality reduction, after which the prediction accuracy remains relatively high with less input data involved (as shown in Fig. 8f). In other words, it assists in focusing the scope of investigation by identifying and prioritizing those variables with higher weights, facilitating the effective identification of predominant variables influencing the mechanical properties of the studied material as demonstrated later. In addition, the maintained high prediction accuracy supports the significance of dimensionality reduction in this work. After obtaining the corresponding weight to each input variable, we can significantly decrease the dimensionality of the training dataset. This approach reduces the effort needed for acquiring certain input variables while maintaining the satisfied mechanical prediction performance.

3.4. PSP relationship development for LPBF AlSi10Mg assisted by interpretable GPR model

The interpretability of GPR model enables the explanation of the input variable weights from the perspective of material science. Overall, weights of each input variable for different mechanical properties have been shown in Fig. 9a. It should be noted that one variable may have varying impact on different mechanical properties. Taking laser power as an example to illustrate its corresponding weight for different mechanical properties (Fig. 9b), yield strength seems to be more sensitive to laser power than ultimate tensile strength in general for LPBF AlSi10Mg. The reason is that process parameters strongly affecting the cooling rate (laser power in this case) will affect the final crystalline lattice defect structure in fabricated AlSi10Mg. During solidification, a higher cooling rate allows less time for dislocations along low-angle grain boundaries to annihilate, resulting in a more distorted lattice with higher kernel average misorientation (KAM) [56]. It comes with more storage of geometrically necessary dislocations (GNDs) that contribute to the strengthening of crystalline materials [57,58]. In other words, the more dense network of dislocations and low angle grain boundaries then act as

barriers to the initial motion of dislocations for higher yield strength. However, the microstructure with a less dense network of dislocations and low angle grain boundaries has a higher capacity for dislocation multiplication and strain hardening, thus giving a similar ultimate tensile strength. Hence, when slightly changing laser power within optimized process window, the UTS is not affected significantly while the yield strength will be affected by the hardening behavior introduced by different KAM as shown in Fig. 9d. Moreover, microstructures tend to have more of an impact on yield strength rather than ductility as shown in Fig. 9c. Due to the correlation between grain size and cell size mentioned above, both grain and cellular microstructures were discussed together here regarding their contribution to mechanical properties. Firstly, the higher strength is attributed to both Hall-Patch strengthening mechanism and the finer cellular structure (Fig. 9e), where more grain boundaries and cellular boundaries (formed by the Si-rich phase) act as barriers for dislocation movement [59]. In the meantime, the typical trade-off of such strengthening mechanisms would be the sacrificing ductility. However, in this case the % elongation is maintained because the higher strength condition also has ~ 0.5 % higher relative density with less manufacturing defects, which

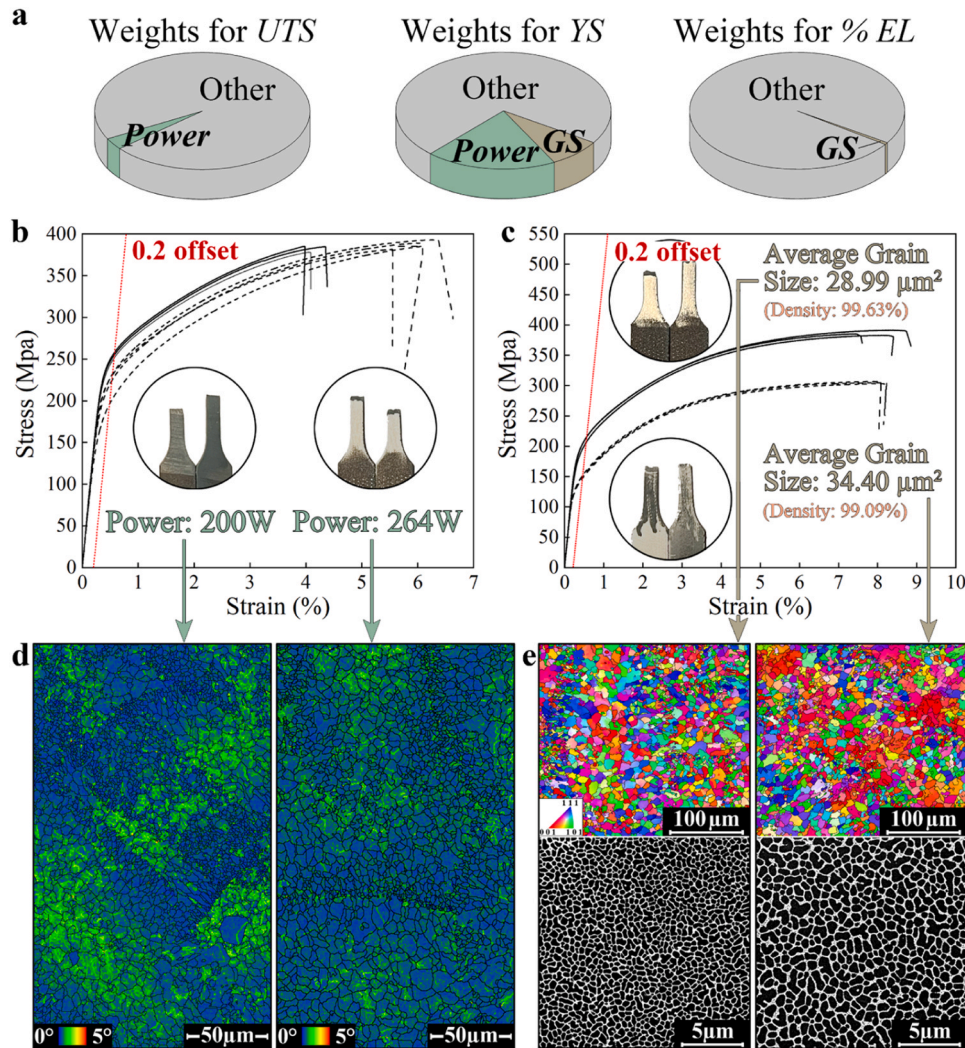


Fig. 9. Interpretability of weights. a, Examples to illustrate the interpretability: weights of laser power (P) and density (ρ) in contributing to three mechanical property predictions in the corresponding single-output models. b, An example of how two laser powers (200 and 264 W with same scan speed for both) may lead to similar ultimate tensile strength but different yield strength. c, An example of two fabricated samples with similar % elongation but different yield strengths. Note: all specimens used here for illustration are outside of the training data scope to avoid overfitting. d, Maps of kernel average misorientation (KAM) of the different fabricated specimens from panel b that illustrate the difference of local misorientation introduced by different laser power. e, Electron back-scattered diffraction (EBSD) and scanning electron microscopy (SEM) images of the different fabricated specimens from panel c, showing the corresponding grains structure and typical Al-Si cellular structure and revealing different microstructure size help to obtain the same ductility with very different yield strengths.

promotes ductility as shown in Fig. 9c. Thus, the final mechanical performance of LPBF AlSi10Mg is affected by multiple factors and cannot be simply explained by a singular linear correlation in its PSP relationship.

Moreover, post-mortem fractography is vital for understanding material failure and serves as a direct verification of mechanical test results. In this study, all LPBF AlSi10Mg specimens display a typical ductile fracture, characterized by the presence of dimple structures on the fracture surface, as shown in Fig. 10. Interestingly, the size of these dimples on fracture surface is similar to the cell size in cellular microstructure of AlSi10Mg, which is consistent with Ref. [60]. The silicon network (Fig. 10c) within the cellular structure acts as a precipitation hardening mechanism and significantly influences the fracture mechanism, affecting the size of dimples as noted in Ref. [42]. For instance, two specimens shown in Fig. 9c demonstrate this relationship: the specimen with smaller cell size forms smaller dimples on the fracture surface (Fig. 10e), whereas the one with larger cell size displays larger dimples (Fig. 10f). Therefore, the correlation between fractography and microstructure not only enhances our understanding but also highlights the potential of integrating fractography data into the training model in future research.

In general, as shown in Fig. 8a, the relative density of the printed sample undoubtedly plays a vital role in the final mechanical performance, regardless of whether it is a single mechanical property prediction or a multi-output model. The interior defect or porosity will lead to stress concentration and provide sites for crack initiation, significantly affecting the mechanical properties LPBF AlSi10Mg. Process parameters (P and v) also significantly affect the model prediction while all mechanical properties are more sensitive to scan speed v . The reason might be that the scan speed directly influences the time of energy exposure to each specific point on the powder bed, thereby having a more immediate effect on the cooling rate. In contrast, while laser power does impact the overall temperature (especially the peak temperature) and thermal gradients, its effect on cooling rate is moderated by the thermal conductivity and specific heat capacity of the material being processed [61]. Moreover, strength (YS and UTS) prediction models have greater weights on melt pool depth, while the tensile elongation (%EL) is more affected by the melt pool width. As mentioned in the Section 2.1, the 90-degree increment scan strategy leads to the criss-cross layered melt pool structure, schematically depicted in Fig. 11g. The applied tensile stress is along the y axis (Fig. 11g), and the fracture surface is

perpendicular to the melt pool layer plane (x - y). In such case, the crack prefers to propagate along the melt pool boundaries [47] (boundaries at the bottom of the melt pools in our case), the size of which is consistent with the general melt pool size affected by the energy density used for manufacturing [60]. Overall, an intermediate melt pool depth is required to provide the best interlayer bonding and ultimate tensile strength (Fig. 11d and g). On the other hand, the ductility of the sample is more affected by the melt pool width. Similar to the depth, the wider melt pool width (i.e., the larger general melt pool size) correlates with a coarser cellular structure at the melt pool boundaries, particularly along the deformation direction in this study (Fig. 11g). Subsequently, the coarse cellular region with larger α -Al grain (compared to hard Si-rich phase) can accommodate more local plastic deformation resulting in better ductility [42,62]. It is worth mentioning that both melt pool width and depth should matter for all mechanical properties, either directly or through correlated microstructural features (e.g., cellular structure), and the different preferred weights can help reveal the corresponding importance for predicting a particular property and underlying physical mechanism of their influence. In addition, melt pool ratio also shows a certain importance in mechanical properties prediction according to its weight. It is well known that a higher melt pool ratio often leads to the keyhole mode [46], subsequently causing the inferior melt pool bonding. This compromised bonding negatively influences the mechanical properties, which is consistent with Ref. [63].

It is also important to note that the influence of each input variable on the mechanical properties is non-monotonic. For instance, lower laser power leading to higher cooling rate and smaller grain size will not always increase the yield strength of the fabricated specimen by fine grain strengthening mechanism; rather, there is a lower limit to laser power where the strength will drop off due to the defects introduced by lack of fusion. Since the final mechanical properties of LPBF AlSi10Mg is affected and determined by multiple factors and involves complex non-linear relationships, it is better to visualize the PSP relationship for LPBF AlSi10Mg from higher dimensionality, i.e., the scope of process window. The effect of different inputs reflected by the weight is beneficial for constructing and understanding PSP relationships of LPBF AlSi10Mg, which is shown in Fig. 11a. It is worth noting that three different colors shown in the figure are used for denoting the three P-S-P portions. However, GPR is not a layered model such as neural networks and it will treat all inputs at the same level [43], as shown in Fig. 4. Taking the

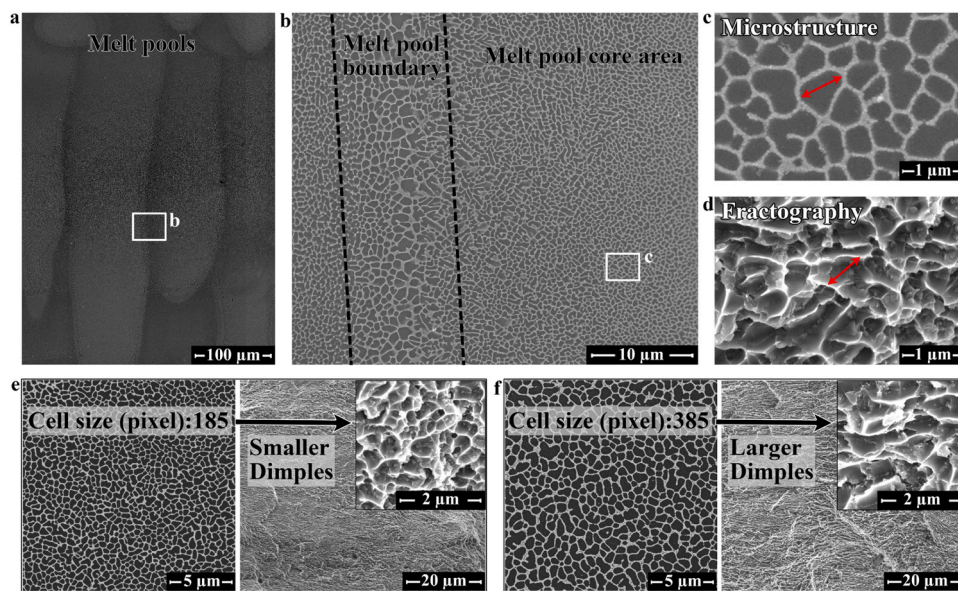


Fig. 10. Fractography and cellular microstructure of LPBF AlSi10Mg. a , melt pool structure of LPBF AlSi10Mg. b, zoomed in SEM images reveals the typical cellular structure of LPBF AlSi10Mg. The size of cells in cellular structure (c) is similar to the size of dimples found on fracture surface (d). e and f, a comparison between two samples in Fig. 9c regarding their cellular structure and fracture surfaces.

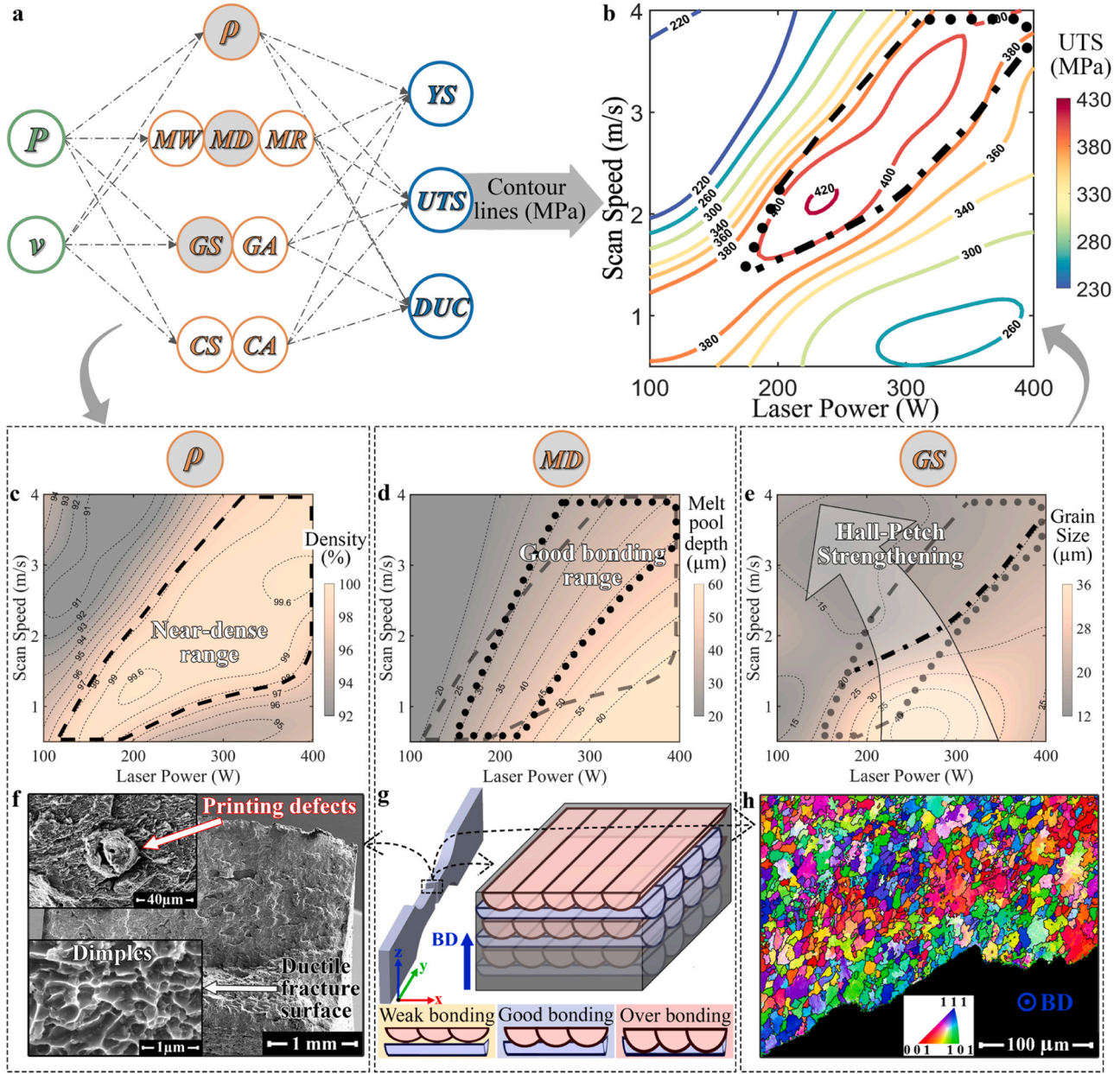


Fig. 11. Interpretable prediction structure. a , Process-structure-property (PSP) relationships for LPBF AlSi10Mg based on the studied variables in this study. b, Contour lines of ultimate tensile strength (UTS) as a function of two process parameters is taken as representative property to illustrate how it is affected by different variables based on interpretable prediction model. The high strength area circled by dash lines can be interpreted based on weights of variables in Fig. 7a. c, Physical density plays a vital role in defining the ultimate tensile strength because failure tends to happen earlier with more printing defects (f). d, Suitable melt pool depth indicates a good bonding structure between layers (g). e, Finer grain size is preferable for better tensile strength with the arrow in the map showing the strengthening trend according to Hall-Petch strengthening principle. h, EBSD image near the tensile fracture surface.

ultimate tensile strength (UTS) as instance, in addition to the process parameters, the relative density and melt pool depth have the most significant impact, followed by the grain size (marked as grey circles in Fig. 11a). Firstly, Fig. 11c marks the suitable process window with > 98 % relative density for LPBF fabricated AlSi10Mg parts. Since defect embedded in the printed parts is usually the initiation position of the crack and detrimentally influence the mechanical performance of the material, the suitable process window targeting relative density is always the first process optimization step in the LPBF material development and investigation process. Such three dimensional printability map where the low energy density (i.e., the combination of low laser power and high scan speed) and high energy density (i.e., the combination of high laser power and low scan speed) areas are associated with the lack

of fusion and key-hole porosity defects, respectively, can also be found from other studies [64,65]. Subsequently, a suitable melt pool depth should be a bit larger than defined layer thickness for good interlayer bonding but not too large triggering key-hole mode and introducing the porosity at the tip of the melt pools. So, a suitable range of melt pool depth ($30 \mu\text{m} < MD < 45 \mu\text{m}$) is marked in Fig. 11d to further refine the process window for desired mechanical performance for LPBF AlSi10Mg. In Fig. 11e, according to Hall-Petch strengthening principle, (i.e., $\Delta\sigma_{\text{Hall-Petch}} = kd^{-1/2}$, where k is a material-dependent constant and d is the grain size), a smaller average grain size is beneficial for promoting dislocation pile-ups at grain boundaries to increase the strength of the material. Working along with finer grain structure, the correlated finer cellular structure (as mentioned above) also aids in localizing and

constraining dislocations within the cells. The white arrow represents the trend of the strengthening effect under the coordinate plane composed of two process parameter variables, where the reduced laser power and increased scan speed result in higher cooling rate, leading to smaller microstructures. In this way, the process-structure-property relationship of LPBF AlSi10Mg can be visualized and interpreted directly. Overall, it can be seen that the process parameters of LPBF directly determines the macro-structure (e.g., printing defects), meso-structure (e.g., melt pool depth) and microstructure (e.g., grain size), which in turn indirectly affect the final mechanical property performance (UTS in this case). The influence of each variable and how the mechanical property vary within the defined process window has been visualized in Fig. 11b.

4. Conclusions

In this study, we have demonstrated a new interpretable approach employing a GPR model to make predictions of mechanical properties for AlSi10Mg fabricated with different LPBF process parameters and previously unexplored process window, the process of which can also be interpreted from the material science perspective to establish the process-structure-property relationship of fabricated metallic materials. The following conclusions can be drawn:

- 1) The multivariate GPR model has great advantages in mechanical property prediction of LPBF AlSi10Mg by involving more variables including process parameters (laser power and scan speed), fabrication defects (relative density), mesostructure (melt pool) and microstructure (grain structure and cellular structure) to build a more robust PSP relationship followed by better prediction accuracy. In addition, our results show that the predictive performance of the model can be further improved when correlating all mechanical properties of fabricated sample applied by multi-output GPR model. In this way, different mechanical properties can be predicted simultaneously as well.
- 2) The prediction error for any targeted input variables generated via GPR model can be used as a guidance for an efficient iterative prediction-experiment-training exploration of the process window for LPBF metallic materials. It can assist in exploring larger process window efficiently (e.g., scenario of higher laser power capacity of the 3D printer) and predicting corresponding mechanical properties when using new fabricating process parameters in unexplored process window area with minimal required experimentation. The larger explored process window can then provide greater potential of the LPBF fabricated materials that gives desirable and previously unattainable mechanical properties.
- 3) The unique weight generation characteristic of GPR provides valuable information for feature selection, which can be used to establish a more efficient prediction model by excluding less important structure variables and reducing input dimensionality. More importantly, the weight of input variables can be beneficial for the establishment of PSP relationships across the defined process window.
- 4) LPBF AlSi10Mg, as illustrated example in this study, has been fully explored regarding optimized process window and the corresponding PSP relationship. Assisted by interpretable machine learning model, the effect of different process parameters and material structures on the final mechanical performance is analyzed and visualized, to better understand the process-structure-property relationship and tailor the mechanical properties for different applications. This approach will be applicable for other AM materials to explore optimized process windows and corresponding PSP relationships.

We believe there is significant potential of the proposed machine learning approach for the development of new AM materials and for exploring process-structure-property relationships. For the application

to new alloys, a few data points can be used for initial training followed by the proposed prediction-experiment process. In our future work, other important process variables (such as hatch space, layer thickness and even post-treatments) or structure variables (such as low/high angle grain boundaries) can also be incorporated into our model to investigate their influence on the structure and properties of the AM materials. Moreover, fractography analysis is able to not only be used as one of the approaches to validate our interpretable model and prediction results, but also be integrated into the model for further investigation. However, it is worth noting that applying the trained model to materials fabricated in a completely different batch (e.g., produced by a different printer or using metal powder from a different supplier) may result in variations in predictions. This is due to the fact that changes in the laser system or powder size distribution could slightly alter the process aspect of the PSP relationship. Therefore, it is crucial to bear in mind that a model trained with a dataset compiled from various sources in the literature might not be as robust in its PSP predictions as one trained with data from consistent material batches.

CRediT authorship contribution statement

Chun H. Wang: Writing – review & editing, Supervision. **Xiaopeng Li:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization. **Vladislav Yakubov:** Writing – review & editing, Writing – original draft, Data curation. **Jamie J. Kruzic:** Writing – review & editing, Supervision. **Qian Liu:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Wenliang Chen:** Writing – review & editing, Data curation.

Declaration of Competing Interest

Authors declare that they have no competing interests.

Data Availability

Data will be made available on request.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.addma.2024.104187](https://doi.org/10.1016/j.addma.2024.104187).

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